

# A candidate 67.302666% lower bound for simple critical-line zeros of the Riemann zeta function

## Abstract

Let  $N(T, 2T)$  count the zeros of the Riemann zeta function with ordinates in  $(T, 2T]$ , with multiplicity, and let  $N_0^s(T, 2T)$  count the simple zeros on the critical line in that interval. This derivative artifact updates the finite certificate in Ainta's AI-assisted draft [2]. Keeping that draft's stability, overlap-kernel, block-aggregation, and pinching arguments, which refine the rank-trace argument of Claude [1], and adding an exact finite-dimensional spectral envelope plus a rigorously reproduced stronger interval certificate gives

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.6730266625438475496579 \dots$$

The finite certificate proves an exact seven-point inequality at the rational target  $382623/10^8$ . Its standalone Arb replay is complete. A substantial Lean formalization has been kernel-checked, but the dependency-closed build of all generated terminal catalogs and the final public theorem is not complete; this paper is therefore released as an AI-assisted, unreviewed research draft, not as a completed formal-verification claim.

## 1 Statement and setup

We use the notation and normalisations of [1, §§2,4,7.1]. Put

$$\ell := \log \frac{T}{2\pi}, \quad I := (T, 2T], \quad D_0 := T^{1/2}, \quad I' := (T - D_0, 2T + D_0].$$

For the optimised test family of Theorem D,  $L = \ell$  and

$$\phi(u) = \cos\left(\frac{\sqrt{2}u}{\ell}\right)^{1/2} \varrho(L/2 - |u|),$$

where the fixed-width ramp  $\varrho$  is defined in [1, §2.2]. Let

$$a := \frac{1}{L} \int_{\mathbb{R}} \phi(u)^2 du, \quad \widehat{G} := \frac{G}{aL^2}, \quad \widehat{A} := \frac{A}{aL^2}, \quad d := \left\lfloor \frac{LT}{2\pi} \right\rfloor.$$

As in [1, §4.1], divide the zeros in  $I'$  into the simple critical-line zeros  $\mathcal{S}_1$ , the distinct multiple critical-line zeros  $\mathcal{S}_2$ , and the unordered off-line pairs  $\mathcal{P}$ . Write

$$s_1 := \#\mathcal{S}_1, \quad s_2 := \#\mathcal{S}_2, \quad p := \#\mathcal{P}.$$

Then

$$N(I') \geq s_1 + 2s_2 + 2p. \tag{1}$$

For  $\rho \in \mathcal{S}_1$ , define

$$u_\rho := (\hat{\phi}(\gamma_\rho - \tau_k))_{0 \leq k < d} \in \mathbb{R}^d, \quad v_\rho := \frac{u_\rho}{\sqrt{a}L}.$$

The simple-zero part of  $\hat{A}$  is

$$P_1 = VV^\top = \sum_{\rho \in \mathcal{S}_1} v_\rho v_\rho^\top,$$

where the columns of  $V$  are the  $v_\rho$ . By Lemma 2.2 of [1],  $\|v_\rho\| \leq 1$ . For

$$Q' := \hat{A} - P_1,$$

the decomposition in the proof of Proposition 4.4 of [1] gives

$$n_+(Q') \leq s_2 + p. \quad (2)$$

Set

$$H_{\text{MT}} := \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 0.6725007036794116457 \dots$$

Theorem D of [1] proves the lower bound  $H_{\text{MT}}$ . The updated finite input gives the following revision of Ainta's bound.

**Theorem 1.1.** *One has*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{864900000000H_{\text{MT}} - 1723600000}{867988393769 - 20000\sqrt{100022498758}} = 0.6730266625438475496579 \dots$$

## 2 A stability refinement of the rank–trace inequality

Define the convex function

$$\Psi(t) := \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t-3, & t \geq 2. \end{cases} \quad (3)$$

For a positive semidefinite matrix  $M$ , let

$$\mathcal{D}(M) := \text{tr } \Psi(M).$$

**Lemma 2.1** (Stability-enhanced rank–trace inequality). *Let  $V$  be a  $d \times r$  matrix whose columns have norm at most one. Put  $P = VV^* \succeq 0$  and  $M = V^*V$ , and let  $Q$  be Hermitian with  $n_+(Q) \leq b$ . Then*

$$\|P + Q\|_{\text{F}}^2 \geq 4 \text{tr}(P + Q) - 3r - 4b + \mathcal{D}(M). \quad (4)$$

*Proof.* Write  $Q = Q_+ - Q_-$  with  $Q_\pm \succeq 0$ ,  $Q_+Q_- = 0$ , and  $\text{rank } Q_+ \leq b$ . Since  $P, Q_+ \succeq 0$ ,

$$\|P + Q\|_{\text{F}}^2 \geq \|P - Q_-\|_{\text{F}}^2 + \|Q_+\|_{\text{F}}^2.$$

If  $q_j > 0$  are the nonzero eigenvalues of  $Q_+$ , then  $q_j^2 \geq 4q_j - 4$ , so

$$\|Q_+\|_{\text{F}}^2 \geq 4 \text{tr } Q_+ - 4b. \quad (5)$$

Let  $p_1 \geq \dots \geq p_r \geq 0$  be the eigenvalues of  $M$  (the nonzero eigenvalues of  $P$ , padded by zeros), and let  $n_1 \geq n_2 \geq \dots \geq 0$  be the eigenvalues of  $Q_-$ . Von Neumann's trace inequality gives  $\text{tr}(PQ_-) \leq \sum_{i \leq r} p_i n_i$ , hence

$$\|P - Q_-\|_{\text{F}}^2 + 4 \text{tr } Q_- \geq \sum_{i \leq r} ((p_i - n_i)^2 + 4n_i).$$

For  $p \geq 0$ ,

$$\min_{n \geq 0} ((p-n)^2 + 4n) = \begin{cases} p^2, & 0 \leq p \leq 2, \\ 4p-4, & p \geq 2, \end{cases} = 2p-1 + \Psi(p).$$

Therefore

$$\|P - Q_-\|_{\mathbb{F}}^2 \geq 2 \operatorname{tr} P - r + \mathcal{D}(M) - 4 \operatorname{tr} Q_-. \quad (6)$$

Combining (5) and (6) gives

$$\|P + Q\|_{\mathbb{F}}^2 \geq 4 \operatorname{tr}(P + Q) - 2 \operatorname{tr} P - r - 4b + \mathcal{D}(M).$$

Since  $\operatorname{tr} P$  is the sum of the squared column norms of  $V$ ,  $\operatorname{tr} P \leq r$ . □

**Corollary 2.2.** *With  $M := V^{\top} V$ ,*

$$s_1 \geq 4 \operatorname{tr} \hat{A} - \|\hat{A}\|_{\mathbb{F}}^2 - 2N(I') + \mathcal{D}(M). \quad (7)$$

*After removing the tails as in Proposition 4.2 of [1],*

$$N_0^s(T, 2T) \geq H_{\text{MT}} N(T, 2T) + \mathcal{D}(M^\circ) - o(N(T, 2T)), \quad (8)$$

*where  $M^\circ$  is the Gram matrix of the retained central simple zeros.*

*Proof.* Apply Lemma 2.1 to  $P_1 + Q' = \hat{A}$ . Equations (1) and (2) give

$$\|\hat{A}\|_{\mathbb{F}}^2 \geq 4 \operatorname{tr} \hat{A} - 3s_1 - 4s_2 - 4p + \mathcal{D}(M) \geq 4 \operatorname{tr} \hat{A} - s_1 - 2N(I') + \mathcal{D}(M),$$

which is (7).

Use the same comparison  $\hat{A} = \hat{G} - \hat{E}$  and tail estimates as in Propositions 4.2 and 4.4 of [1]. Pinching the full Gram matrix into the retained central block and its complement gives  $\mathcal{D}(M) \geq \mathcal{D}(M^\circ)$  because  $X \mapsto \operatorname{tr} \Psi(X)$  is convex and unitarily invariant. Moreover  $N(I') = N(T, 2T) + o(N(T, 2T))$ , and Theorem D gives

$$\operatorname{tr} \hat{G} = N(T, 2T)(1 + o(1)), \quad \|\hat{G}\|_{\mathbb{F}}^2 = \left( \frac{1}{c_1^*} + o(1) \right) N(T, 2T),$$

where  $2 - 1/c_1^* = H_{\text{MT}}$ . Substitution in (7) yields (8). □

### 3 The Montgomery–Taylor overlap kernel

Put  $h := 2\pi/L$  and define the normalised ordinate

$$x_\rho := \frac{\gamma_\rho - T}{h} = \frac{L(\gamma_\rho - T)}{2\pi}.$$

**Lemma 3.1.** *Fix  $R_0 > 0$ . Delete the simple zeros in strips of normalised width  $L^2$  at the two ends of the grid. The number deleted is  $o(N(T, 2T))$ , and uniformly for retained simple zeros with  $|x_\rho - x_{\rho'}| \leq R_0$ ,*

$$\langle v_\rho, v_{\rho'} \rangle = k(x_\rho - x_{\rho'}) + o(1), \quad (9)$$

where

$$k(x) := \frac{K(x)}{K(0)}, \quad (10)$$

$$\begin{aligned} K(x) &:= \int_{-1/2}^{1/2} \cos(\sqrt{2}t) \cos(2\pi xt) dt \\ &= \frac{\sin(\pi x - 1/\sqrt{2})}{2\pi x - \sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{2\pi x + \sqrt{2}}, \end{aligned} \quad (11)$$

and  $K(0) = \sqrt{2} \sin(1/\sqrt{2})$ . The singularities in (11) are removable.

*Proof.* For the full Gabor grid, Lemma 2.2 of [1] gives

$$\frac{1}{aL^2} \sum_{j \in \mathbb{Z}} \widehat{\phi}(\gamma - \tau_j) \widehat{\phi}(\gamma' - \tau_j) = \frac{\Phi(\gamma - \gamma')}{aL}, \quad \Phi = \widehat{\phi^2}.$$

For points at normalised distance at least  $L^2$  from the grid ends, the terms with  $j \notin [0, d]$  contribute  $O(L^{-2})$ , uniformly when  $|x_\rho - x_{\rho'}|$  is bounded, by the  $r^{-2}$  decay used in [1, §5.3].

For fixed  $x$ , substitution  $u = Lt$  gives the normalised full-grid overlap

$$\frac{\Phi(hx)}{aL} = \frac{\int_{-1/2}^{1/2} \phi(Lt)^2 e^{-2\pi i x t} dt}{\int_{-1/2}^{1/2} \phi(Lt)^2 dt} \rightarrow \frac{K(x)}{K(0)}. \quad (12)$$

Indeed,  $\phi(Lt)^2 \rightarrow \cos(\sqrt{2}t) \mathbf{1}_{[-1/2, 1/2]}(t)$  in  $L^1$ ; the fixed-width taper has rescaled length  $O(L^{-1})$ . The convergence in (12) is uniform for  $x$  in compact sets. Together with the tail estimate, this proves (9). Finally, the deleted strips have ordinate length  $O(L)$  and contain  $O(L^2) = o(N(T, 2T))$  zeros, since  $N(t+1) - N(t) \ll \log(t+3)$ .  $\square$

Set

$$w(x) := k(x)^2.$$

## 4 A seven-point local inequality

For nonnegative gaps  $g_1, \dots, g_6$ , define

$$\mathcal{F}_6(g_1, \dots, g_6) := \frac{1}{3000} \sum_{i=1}^6 g_i + \sum_{r=1}^6 \frac{2}{7-r} \sum_{i=1}^{7-r} w(g_i + \dots + g_{i+r-1}). \quad (13)$$

The second term contains all 21 pairwise separations among seven ordered points.

**Proposition 4.1** (Certified local inequality). *For every  $g_1, \dots, g_6 \geq 0$ ,*

$$\mathcal{F}_6(g_1, \dots, g_6) \geq \frac{382623}{100000000}. \quad (14)$$

*Verification.* The accompanying verifier derives from the public implementation in [2] and uses Arb interval arithmetic through `python-flint`. It evaluates the entire-sinc expression

$$K(x) = \frac{1}{2} \left( \frac{\sin(\pi x - 1/\sqrt{2})}{\pi x - 1/\sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{\pi x + 1/\sqrt{2}} \right)$$

at 128-bit precision and constructs rigorous lower bounds for  $w$  on a grid of mesh  $1/4000$ . If the linear pressure alone reaches the target, the box is immediately closed. On the remaining compact region, the one-variable term

$$U(g) := \frac{g}{3000} + \frac{1}{3}w(g)$$

reduces each coordinate to certified unions of grid cells. The resulting boxes are proved by interval lower bounds, certified tangent and convexity bounds, a bounded subcell tangent tree for the hardest cases, or bisection of the widest coordinate. The program fails if a terminal box remains unresolved.

The committed certificate records `verified=true`, grid 4000, precision 128 bits, 980,069 visited nodes, 490,399 pruned nodes, 489,670 splits, and maximum depth 65. The terminal breakdown is 285,258 interval, 3,166 pressure, and 201,975 tangent terminals; 1,773 of the tangent terminals use the bounded subcell fallback. The report also records the hashes of the tables and certificate data used in the search.  $\square$

**Lemma 4.2.** *Let  $y_1 < \dots < y_m$  with  $m \geq 7$ , and put*

$$E_m := 2 \sum_{1 \leq i < j \leq m} w(y_j - y_i).$$

*Then*

$$E_m + \frac{1}{500}(y_m - y_1) \geq \frac{382623}{100000000}(m - 6). \quad (15)$$

*Proof.* Write  $g_i = y_{i+1} - y_i$  and sum (14) over the  $m - 6$  consecutive seven-point windows. A pair spanning  $r$  gaps occurs at most  $7 - r$  times and has coefficient  $2/(7 - r)$  each time, so its total contribution is at most  $2w(y_j - y_i)$ . Each single gap occurs at most six times, giving at most

$$\frac{6}{3000} \sum_{i=1}^{m-1} g_i = \frac{1}{500}(y_m - y_1).$$

$\square$

**Lemma 4.3** (Exact spectral envelope). *Let  $G$  be an  $m \times m$  correlation matrix, and put*

$$E := \text{tr}(G - I)^2 = 2 \sum_{i < j} |G_{ij}|^2, \quad D := \text{tr} \Psi(G).$$

*For  $0 \leq E \leq m(m - 1)$ , define*

$$g_m(E) := \begin{cases} E, & 0 \leq E \leq m/(m - 1), \\ E/m + 2\sqrt{(m - 1)E/m} - 1, & m/(m - 1) \leq E \leq m(m - 1). \end{cases} \quad (16)$$

*Then  $D \geq g_m(E)$ . Moreover,  $g_m$  is increasing and 1-Lipschitz.*

*Proof.* Write the eigenvalues of  $G$  as  $1 + y_i$ . Then  $y_i \geq -1$ ,  $\sum_i y_i = 0$ , and  $E = \sum_i y_i^2$ . If every  $y_i \leq 1$ , then  $D = E$ . Otherwise let  $H = \{i : y_i > 1\}$ , put  $h = |H|$ , write  $y_i = 1 + u_i$  on  $H$ , and set

$$q_+ = \sum_{i \in H} u_i, \quad r^2 = \sum_{i \in H} u_i^2, \quad R = \sum_{i \notin H} y_i^2.$$

The complementary deviations sum to  $-(h + q_+)$ , so Cauchy's inequality and  $q_+ \geq r$  give

$$\begin{aligned} E &\geq h + 2r + r^2 + \frac{(h + r)^2}{m - h} \\ &= \frac{m(1 + r)^2}{m - 1} + \frac{(h - 1)(m + r)^2}{(m - h)(m - 1)}. \end{aligned}$$

Since  $E - D = r^2$ , this implies

$$D \geq E/m + 2\sqrt{(m - 1)E/m} - 1.$$

It also shows that the second case requires  $E > m/(m - 1)$ . Equality is realized by the correlation matrix

$$\left(1 - \frac{a}{m - 1}\right)I + \frac{a}{m - 1}J, \quad a = \sqrt{(m - 1)E/m}.$$

On the second branch,

$$g'_m(E) = \frac{1}{m} + \sqrt{\frac{m - 1}{mE}} \leq 1,$$

which also proves monotonicity and the Lipschitz assertion.  $\square$

Take

$$m := 279, \quad A := \frac{382623}{100000000}(m - 6) = \frac{104456079}{100000000}, \quad (17)$$

and put

$$C := g_{279}(A) = \frac{11606231}{3100000000} + 2\sqrt{\frac{1613266109}{1550000000}} - 1. \quad (18)$$

If  $e + x \geq A$ , monotonicity and the 1-Lipschitz property imply

$$g_m(e) + x \geq g_m(A) = C. \quad (19)$$

Indeed, this is immediate for  $e \geq A$ , while for  $e < A$  one has  $x \geq A - e \geq g_m(A) - g_m(e)$ .

Let  $B$  be a block of 279 consecutive retained simple zeros, and write  $G_B$  for its Gram matrix. If its span term already exceeds  $A$ , the following estimate is immediate because  $C \leq A$ . Otherwise the span is bounded, so Lemma 3.1 applies uniformly to every pair. The diagonal entries are  $1 + o(1)$ ; normalizing the diagonal produces a correlation matrix that differs from  $G_B$  by  $o(1)$  in Frobenius norm. Since  $m$  is fixed and  $\Psi$  is globally 2-Lipschitz, this changes both the defect and the off-diagonal energy by  $o(1)$ . Lemma 4.2, Lemma 4.3, and (19) therefore give uniformly in  $B$

$$\mathcal{D}(G_B) + \frac{1}{500} \text{span}(B) \geq C - o(1). \quad (20)$$

## 5 Shifted block pinching

Let

$$x_1 < \cdots < x_{S^\circ}$$

be the normalised ordinates of the retained central simple zeros. Then

$$S^\circ = N_0^s(T, 2T) - o(N(T, 2T)). \quad (21)$$

For each of the  $m = 279$  offsets, partition the ordered points into full consecutive blocks of size  $m$ , leaving at most  $2(m - 1)$  endpoint points. Pinching  $M^\circ$  to the full principal blocks and the remaining block gives

$$\mathcal{D}(M^\circ) \geq \sum_B \mathcal{D}(G_B). \quad (22)$$

This follows because pinching is an average of unitary conjugations and  $X \mapsto \text{tr} \Psi(X)$  is convex and unitarily invariant.

For each offset, sum (20) over the full blocks and use (22); then average over the  $m$  offsets. The average number of full blocks is  $S^\circ/m + O(1)$ . Each gap  $x_{j+1} - x_j$  belongs to a block span for at most  $m - 1$  offsets, and

$$x_{S^\circ} - x_1 \leq \frac{T}{h} = \frac{LT}{2\pi} = d + O(1) = N(T, 2T) + o(N(T, 2T))$$

by the Riemann–von Mangoldt formula. The blockwise  $o(1)$  term in (20) is uniform, so its total is  $o(N(T, 2T))$ . Therefore

$$\mathcal{D}(M^\circ) \geq \frac{C}{m} N_0^s(T, 2T) - \frac{m-1}{500m} N(T, 2T) - o(N(T, 2T)). \quad (23)$$

With  $m = 279$ ,

$$\mathcal{D}(M^\circ) \geq \frac{C}{279} N_0^s(T, 2T) - \frac{139}{69750} N(T, 2T) - o(N(T, 2T)). \quad (24)$$

Substitute (24) into (8). This gives

$$\begin{aligned} N_0^s(T, 2T) &\geq H_{\text{MT}} N(T, 2T) + \frac{C}{279} N_0^s(T, 2T) \\ &\quad - \frac{139}{69750} N(T, 2T) - o(N(T, 2T)). \end{aligned}$$

Hence

$$\left(1 - \frac{C}{279}\right) N_0^s(T, 2T) \geq \left(H_{\text{MT}} - \frac{139}{69750}\right) N(T, 2T) - o(N(T, 2T)).$$

Divide by  $N(T, 2T)$  and let  $T \rightarrow \infty$  to obtain

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{H_{\text{MT}} - 139/69750}{1 - C/279} = 0.6730266625438475496579 \dots$$

Clearing rational factors gives the exact expression stated in Theorem 1.1. Directed 256-bit Arb arithmetic separates this value between decimals ending in  $\dots 0978$  and  $\dots 0979$ . An exact scan through  $7 \leq m \leq 1000$  gives a unique maximum at  $m = 279$ , with  $m = 278$  second. For completeness, the infinite tail is controlled as follows. Put  $q = 382623/10^8$  and, for real  $x \geq 1000$ , set

$$s(x) := \sqrt{q \left(x - 7 + \frac{6}{x}\right)}.$$

For the continuous version of the second spectral branch,

$$\frac{C_x}{x} = \frac{q(x-6)}{x^2} + \frac{2s(x)-1}{x}.$$

The first term decreases for  $x > 12$ . The derivative of the second term has the sign of

$$s(x) - q \left(x - 14 + \frac{18}{x}\right).$$

At  $x = 1000$ , exact rational arithmetic gives

$$q(1000 - 14)^2 - 1000 = \frac{67996137527}{25000000} > 0, \quad 2q(1000 - 14) - 1 = \frac{163633139}{25000000} > 0.$$

Consequently  $q(x - 14)^2 - x$  is positive and increasing for  $x \geq 1000$ , and

$$s(x) < \sqrt{qx} < q(x - 14) < q\left(x - 14 + \frac{18}{x}\right).$$

Thus  $C_x/x$  decreases. The numerator  $H_{\text{MT}} - 1/500 + 1/(500x)$  also decreases, while the positive denominator  $1 - C_x/x$  increases. Hence  $B_x$  decreases for  $x \geq 1000$ , making  $m = 279$  the unique optimum over every integer  $m \geq 7$ . This proves Theorem 1.1.

## Reproducibility

Proposition 4.1 is the finite computer-assisted input. The accompanying artifact contains its source, tests, trust-base description, and certificate. The verifier reconstructs every transcendental interval enclosure from the stated formulas; a successful replay reports the exact target  $382623/100000000$  and `verified=true`.

The artifact also contains source and partially completed kernel evidence for a Lean certificate consumer. At the snapshot date, the generic MVT, segment, replay, and scaled-checker semantics and the complete code-9 catalog have been kernel-built; the code-10 production catalog is in progress, and the full code-2 catalog and final dependency-closed public build are deferred. Accordingly, this draft does not claim a completed transitive Lean verification. The machine-readable status matrix and the final gates are recorded in `VERIFICATION_STATUS.md`.

This work is AI-assisted and unreviewed. It is a derivative of the earlier public draft [2]; the stability, overlap-kernel, block-aggregation, and shifted-pinching arguments originate there. The new contribution in this revision is the exact finite-dimensional spectral envelope, the stronger exact finite certificate and its reproducibility package, and the partially completed Lean certificate-consumer effort. The clean-room reproduction [3] provides a separate comparison point. The numerical statement should be described as a candidate improvement within this certificate lineage, not as an established record, until independent review has taken place.

## References

- [1] Claude, *More than two thirds of the zeros of the Riemann zeta function lie on the critical line*, preprint, August 10, 2026, available online.
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